

Roll No.

TBT-702

**B. Tech. (BIOTECHNOLOGY)
(SEVENTH SEMESTER)**

MID SEMESTER EXAMINATION, 2019

**INTELLECTUAL PROPERTY RIGHTS,
BIOSAFETY AND BIOETHICS**

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Explain the issues concerning biopiracy and bioprospecting.

(b) What are the remedies to check copyright infringement ?

(c) Write short notes on the following :

(i) PCT

(ii) WIPO

(2) TBT-702

2. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)
- (a) Discuss the importance and types of trademarks.
 - (b) Write a note on patent infringement and remedies.
 - (c) Comment on the following :
 - (i) TKDLs
 - (ii) Farmers Rights
3. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)
- (a) Elaborate on the structure and functions of WTG.
 - (b) Comment on the scope of protection under the Design Act.
 - (c) Give a case study on the patenting of micro-organisms.
4. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)
- (a) Explain PVPA with the help of a case study.
 - (b) Comment on the following :
 - (i) Sui generis system.
 - (ii) Difference between assignment and license of copyright.
 - (c) Explain the Budapest Treaty.

(3)

5. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)
- (a) Comment on the TRIPs and GATT agreement and its relevance with respect to India.
 - (b) Discuss the procedure for obtaining patents using appropriate flowchart.
 - (c) Explain the criteria of patentability and the patentable subject-matter in the biotechnology domain.

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TBT-704

**B. TECH. (BIOTECHNOLOGY)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019**

STEM CELL TECHNOLOGY

Time : 1 : 30 Hours

Maximum Marks : 50

Note :(i) All questions are compulsory.

(ii) Answer any *two* sub questions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) What are stem cells ? Discuss different types of stem cells along with their properties.

(b) Describe potency in context of stem cells and name different types of stem cells based on potency.

- (c) (i) Compare and differentiate between embryonic stem cells and adult stem cells.

(ii) Differentiate between somatic cells and stem cells.

2. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)

- (a) What are different sources of stem cell ?
How are they procured and propagated ?
- (b) Discuss the ethical issues involved embryonic stem cell research.

- (c) Write brief notes on any *two* of the following :

- (i) Wnt signalling
- (ii) Notch-Lin signalling
- (iii) Blastocyst
- (iv) Three germ layers

3. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)

- (a) Describe SCNT. What is its relevance to biotechnology ?

- (b) How can you cure a patient suffering from leukemia using stem cells ?

- (c) What is fate mapping ? Describe different methods of fate mapping and their uses.

4. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)

- (a) What is cryopreservation ? How is it relevant to stem cell culture ?

- (b) Differentiate and compare SCNT and handmade cloning. Which one is advantageous ?

- (c) Describe different tests for pluripotency.

5. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)

- (a) Discuss stem cell patterning and differentiation.

- (b) How would isolate and culture stem cells from a somatic cell ?

- (c) Write short notes on any *two* of the following :

- (i) Gastrula
- (ii) Epigenetics
- (iii) Cellular reprogramming
- (iv) Differential gene expression

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TBT-705

B. Tech. (BIOTECHNOLOGY)

(SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, 2019

PYTHON

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) What is an interpreter ? Compare compiler with interpreter.
- (b) What is Python ? What are the benefits of using Python ?
- (c) What is a statement in a Python Program ? Give *two* examples.

2. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)

(a) What is wrong with the given statement that attempts to assign the value ten to variable x ? $10 = x$

(b) Write the output of the following code fragment :

p = 2

q = 2

p + = 1

q - = 1

print(p)

print(q)

(c) Write a Python program to calculate and print the Area of a Circle. Input the radius from the keyboard.

3. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)

(a) Write a Python program to input a Person Name and age from keyboard and print whether the person "Name:" is eligible to vote or not ?

(b) Write a Python program to check whether a number is Prime or not ?

(c) What is a Function ? Write a Python function to print first 10 numbers of Fibonacci Series.

4. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)

(a) Explain the use of break and continue statements in Python with suitable program.

(b) How are arguments passed by value or by reference ?

(c) Explain the Immutable built-in types in Python.

5. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)

(a) What are the Built-in Data Types in Python ? Explain the scope and use of each type.

(b) How are arguments passed in Python by value or by reference ? Explain.

(c) What is negative index in Python ? How can you convert a number to a string ?